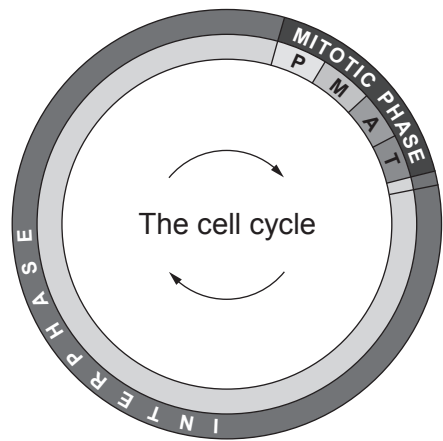


7. A eukaryotic cell cycle is represented in the **Image 7.1**.

Image 7.1



When looking at a group of cells under the microscope, cells undergoing mitosis can be seen as they will have visible chromosomes. The mitotic index, which represents the proportion of cells undergoing mitosis within a group of cells, can be calculated as follows:

$$\text{Mitotic index} = \frac{\text{cells in mitosis}}{\text{total number of cells in field of view}}$$

Cyclins are a family of regulatory proteins that control the progression of the cell cycle. The concentration of cyclin B peaks at the start of mitosis before dropping rapidly as the cells exit mitosis. Some mutations result in abnormally high concentrations of cyclin B.

Describe the events occurring during interphase that are necessary before mitosis can occur.

Using examples, explain why some tissues in humans naturally have a higher mitotic index than others.

Suggest and explain the possible effects of abnormally high concentrations of cyclin B. [9 QER]

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